Student Project Portfolio & Showcase App

UML Component Architecture | Shashank Arya | PES1UG24CS432 | Problem Statement #04

PRESENTATION LAYER

Showcase Portal

Presentation boundary

Provides -

Requires IShowcaseCatalog, IProjectSubmission, IRubricEvaluation, IPublicVoting, ILeaderboard

IShowcaseCatalog

IProjectSubmission

IRubricEvaluation

APPLICATION / BUSINESS LAYER

Showcase Profile

Team profile and published project catalog

Provides IShowcaseCatalog

Requires IProjectData

IShowcaseCatalog

Project Submission

Upload workflow and artifact references

Provides IProjectSubmission

Requires IShowcaseCatalog, IProjectData, IArtifactStore

Evaluation & Rubric

Judge scores, comments, and results

Provides IRubricEvaluation

Requires IShowcaseCatalog, IProjectData

Identity Verification

· voter and returns verified reference

Provides IIdentityVerification

Requires -

IPublicVoting

IIdentityVerification

Public Voting

Authorizes vote and enforces one per project

Provides IPublicVoting, IAcceptedVoteEvents

Requires IIdentityVerification, IShowcaseCatalog, IVoteLedger

IAcceptedVoteEvents

Leaderboard & Realtime

Aggregates counts and publishes ranking updates

Provides ILeaderboard

Requires IShowcaseCatalog, IVoteLedger, IAcceptedVoteEvents

ILeaderboard

DATA LAYER

IVoteLedger

IProjectData

IArtifactStore

Project Data Repository

Profiles, projects, artifact refs, evaluations

Provides IProjectData

Requires -

Vote Ledger

Atomic unique voter/project vote records

Provides IVoteLedger

Requires -

Multimedia Artifact Store

Uploaded media objects and artifact references

Provides IArtifactStore

Requires -

Assembly connectors pair a required socket at the consumer with a provided lollipop at the provider.

Vote flow: identity verification -> atomic unique (verifiedVoterRef, projectId) commit -> VoteAccepted event -> refreshed live ranking.

Ledger uniqueness supports one vote per project; identity verification supports NFR-002 but does not guarantee elimination of every sybil attack.